

Computação Quântica

Prof. Wilson Rabelo
ITEC - UFPA

Álgebra linear: Uma revisão de conceitos básicos para a computação quântica

Cap.1 - Fundamentos de Mecânica Quântica

- Álgebra Linear: Uma revisão de conceitos básicos

The basic objects of linear algebra are *vector spaces*. The vector space of most interest to us is $\mathbf{C}^n$, the space of all n -tuples of complex numbers, $(z_1, \dots, z_n)$. The elements of a vector space are called *vectors*, and we will sometimes use the column matrix notation

$$\begin{bmatrix} z_1 \\ \vdots \\ z_n \end{bmatrix} \quad (2.1)$$

to indicate a vector. There is an *addition* operation defined which takes pairs of vectors to other vectors. In $\mathbf{C}^n$ the addition operation for vectors is defined by

$$\begin{bmatrix} z_1 \\ \vdots \\ z_n \end{bmatrix} + \begin{bmatrix} z'_1 \\ \vdots \\ z'_n \end{bmatrix} \equiv \begin{bmatrix} z_1 + z'_1 \\ \vdots \\ z_n + z'_n \end{bmatrix}, \quad (2.2)$$

where the addition operations on the right are just ordinary additions of complex numbers. Furthermore, in a vector space there is a *multiplication by a scalar* operation. In $\mathbb{C}^n$ this operation is defined by

$$z \begin{bmatrix} z_1 \\ \vdots \\ z_n \end{bmatrix} \equiv \begin{bmatrix} zz_1 \\ \vdots \\ zz_n \end{bmatrix}, \quad (2.3)$$

Quantum mechanics is our main motivation for studying linear algebra, so we will use the standard notation of quantum mechanics for linear algebraic concepts. The standard quantum mechanical notation for a vector in a vector space is the following:

$$|\psi\rangle. \quad (2.4)$$

Notation	Description
z^*	Complex conjugate of the complex number z . $(1 + i)^* = 1 - i$
$ \psi\rangle$	Vector. Also known as a <i>ket</i> .
$\langle\psi $	Vector dual to $ \psi\rangle$. Also known as a <i>bra</i> .
$\langle\varphi \psi\rangle$	Inner product between the vectors $ \varphi\rangle$ and $ \psi\rangle$.
$ \varphi\rangle \otimes \psi\rangle$	Tensor product of $ \varphi\rangle$ and $ \psi\rangle$.
$ \varphi\rangle \psi\rangle$	Abbreviated notation for tensor product of $ \varphi\rangle$ and $ \psi\rangle$.
A^*	Complex conjugate of the A matrix.
A^T	Transpose of the A matrix.
$A^\dagger$	Hermitian conjugate or adjoint of the A matrix, $A^\dagger = (A^T)^*$. $\begin{bmatrix} a & b \\ c & d \end{bmatrix}^\dagger = \begin{bmatrix} a^* & c^* \\ b^* & d^* \end{bmatrix}.$
$\langle\varphi A \psi\rangle$	Inner product between $ \varphi\rangle$ and $A \psi\rangle$. Equivalently, inner product between $A^\dagger \varphi\rangle$ and $ \psi\rangle$.

Figure 2.1. Summary of some standard quantum mechanical notation for notions from linear algebra. This style of notation is known as the *Dirac* notation.

2.1.1 Bases and linear independence

A *spanning set* for a vector space is a set of vectors $|v_1\rangle, \dots, |v_n\rangle$ such that any vector $|v\rangle$ in the vector space can be written as a linear combination $|v\rangle = \sum_i a_i |v_i\rangle$ of vectors

in that set. For example, a spanning set for the vector space $\mathbf{C}^2$ is the set

$$|v_1\rangle \equiv \begin{bmatrix} 1 \\ 0 \end{bmatrix}; \quad |v_2\rangle \equiv \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad (2.5)$$

since any vector

$$|v\rangle = \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} \quad (2.6)$$

in $\mathbf{C}^2$ can be written as a linear combination $|v\rangle = a_1|v_1\rangle + a_2|v_2\rangle$ of the vectors $|v_1\rangle$ and $|v_2\rangle$. We say that the vectors $|v_1\rangle$ and $|v_2\rangle$ *span* the vector space $\mathbf{C}^2$.

Generally, a vector space may have many different spanning sets. A second spanning set for the vector space $\mathbf{C}^2$ is the set

$$|v_1\rangle \equiv \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}; \quad |v_2\rangle \equiv \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \quad (2.7)$$

since an arbitrary vector $|v\rangle = (a_1, a_2)$ can be written as a linear combination of $|v_1\rangle$ and $|v_2\rangle$,

$$|v\rangle = \frac{a_1 + a_2}{\sqrt{2}}|v_1\rangle + \frac{a_1 - a_2}{\sqrt{2}}|v_2\rangle. \quad (2.8)$$

A set of non-zero vectors $|v_1\rangle, \dots, |v_n\rangle$ are *linearly dependent* if there exists a set of complex numbers $a_1, \dots, a_n$ with $a_i \neq 0$ for at least one value of i , such that

$$a_1|v_1\rangle + a_2|v_2\rangle + \dots + a_n|v_n\rangle = 0. \quad (2.9)$$

A set of vectors is *linearly independent* if it is not linearly dependent. It can be shown that any two sets of linearly independent vectors which span a vector space V contain the same number of elements. We call such a set a *basis* for V . Furthermore, such a basis set always exists. The number of elements in the basis is defined to be the *dimension* of V . In this book we will only be interested in *finite dimensional* vector spaces. There are

Exercise 2.1: (Linear dependence: example) Show that $(1, -1)$, $(1, 2)$ and $(2, 1)$ are linearly dependent.

2.1.2 Linear operators and matrices

A *linear operator* between vector spaces V and W is defined to be any function $A : V \rightarrow W$ which is linear in its inputs,

$$A \left(\sum_i a_i |v_i\rangle \right) = \sum_i a_i A(|v_i\rangle) . \quad (2.10)$$

Usually we just write $A|v\rangle$ to denote $A(|v\rangle)$. When we say that a linear operator A is defined *on* a vector space, V , we mean that A is a linear operator from V to V . An important linear operator on any vector space V is the *identity operator*, I_V , defined by the equation $I_V|v\rangle \equiv |v\rangle$ for all vectors $|v\rangle$. Where no chance of confusion arises we drop the subscript V and just write I to denote the identity operator. Another important linear operator is the *zero operator*, which we denote 0 . The zero operator maps all vectors to

the zero vector, $0|v\rangle \equiv 0$. It is clear from (2.10) that once the action of a linear operator A on a basis is specified, the action of A is completely determined on all inputs.

Suppose V, W , and X are vector spaces, and $A : V \rightarrow W$ and $B : W \rightarrow X$ are linear operators. Then we use the notation BA to denote the *composition* of B with A , defined by $(BA)(|v\rangle) \equiv B(A(|v\rangle))$. Once again, we write $BA|v\rangle$ as an abbreviation for $(BA)(|v\rangle)$.

The most convenient way to understand linear operators is in terms of their *matrix representations*. In fact, the linear operator and matrix viewpoints turn out to be completely equivalent. The matrix viewpoint may be more familiar to you, however. To see the connection, it helps to first understand that an m by n complex matrix A with entries A_{ij} is in fact a linear operator sending vectors in the vector space $\mathbf{C}^n$ to the vector space $\mathbf{C}^m$, under matrix multiplication of the matrix A by a vector in $\mathbf{C}^n$. More precisely, the claim that the matrix A is a linear operator just means that

$$A \left(\sum_i a_i |v_i\rangle \right) = \sum_i a_i A|v_i\rangle \quad (2.11)$$

We've seen that matrices can be regarded as linear operators. Can linear operators be given a matrix representation? In fact they can, as we now explain. This equivalence between the two viewpoints justifies our interchanging terms from matrix theory and operator theory throughout the book. Suppose $A : V \rightarrow W$ is a linear operator between vector spaces V and W . Suppose $|v_1\rangle, \dots, |v_m\rangle$ is a basis for V and $|w_1\rangle, \dots, |w_n\rangle$ is a basis for W . Then for each j in the range $1, \dots, m$, there exist complex numbers A_{1j} through A_{nj} such that

$$A|v_j\rangle = \sum_i A_{ij}|w_i\rangle. \quad (2.12)$$

The matrix whose entries are the values A_{ij} is said to form a *matrix representation* of the operator A . This matrix representation of A is completely equivalent to the operator A , and we will use the matrix representation and abstract operator viewpoints interchangeably. Note that to make the connection between matrices and linear operators we must specify a set of input and output basis states for the input and output vector spaces of the linear operator.

Exercise 2.2: (Matrix representations: example) Suppose V is a vector space with basis vectors $|0\rangle$ and $|1\rangle$, and A is a linear operator from V to V such that $A|0\rangle = |1\rangle$ and $A|1\rangle = |0\rangle$. Give a matrix representation for A , with respect to the input basis $|0\rangle, |1\rangle$, and the output basis $|0\rangle, |1\rangle$. Find input and output bases which give rise to a different matrix representation of A .

Exercise 2.3: (Matrix representation for operator products) Suppose A is a linear operator from vector space V to vector space W , and B is a linear operator from vector space W to vector space X . Let $|v_i\rangle, |w_j\rangle$, and $|x_k\rangle$ be bases for the vector spaces V, W , and X , respectively. Show that the matrix representation for the linear transformation BA is the matrix product of the matrix representations for B and A , with respect to the appropriate bases.

Exercise 2.4: (Matrix representation for identity) Show that the identity operator on a vector space V has a matrix representation which is one along the diagonal and zero everywhere else, if the matrix representation is taken with respect to the same input and output bases. This matrix is known as the *identity matrix*.

2.1.3 The Pauli matrices

Four extremely useful matrices which we shall often have occasion to use are the *Pauli matrices*. These are 2 by 2 matrices, which go by a variety of notations. The matrices, and their corresponding notations, are depicted in Figure 2.2. The Pauli matrices are so useful in the study of quantum computation and quantum information that we encourage you to memorize them by working through in detail the many examples and exercises based upon them in subsequent sections.

$$\begin{aligned}\sigma_0 \equiv I &\equiv \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} & \sigma_1 \equiv \sigma_x \equiv X &\equiv \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \\ \sigma_2 \equiv \sigma_y \equiv Y &\equiv \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} & \sigma_3 \equiv \sigma_z \equiv Z &\equiv \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}\end{aligned}$$

Figure 2.2. The Pauli matrices. Sometimes I is omitted from the list with just X, Y and Z known as the Pauli matrices.

2.1.4 Inner products

An *inner product* is a function which takes as input two vectors $|v\rangle$ and $|w\rangle$ from a vector space and produces a complex number as output. For the time being, it will be convenient to write the inner product of $|v\rangle$ and $|w\rangle$ as $(|v\rangle, |w\rangle)$. This is not the standard quantum mechanical notation; for pedagogical clarity the $(\cdot, \cdot)$ notation will be useful occasionally in this chapter. The standard quantum mechanical notation for the inner product $(|v\rangle, |w\rangle)$ is $\langle v|w\rangle$, where $|v\rangle$ and $|w\rangle$ are vectors in the inner product space, and the notation $\langle v|$ is used for the *dual vector* to the vector $|v\rangle$; the dual is a linear operator from the inner product space V to the complex numbers $\mathbf{C}$, defined by $\langle v|(|w\rangle) \equiv \langle v|w\rangle \equiv (|v\rangle, |w\rangle)$.

A function $(\cdot, \cdot)$ from $V \times V$ to $\mathbf{C}$ is an inner product if it satisfies the requirements that:

(1) $(\cdot, \cdot)$ is linear in the second argument,

$$\left(|v\rangle, \sum_i \lambda_i |w_i\rangle \right) = \sum_i \lambda_i (|v\rangle, |w_i\rangle). \quad (2.13)$$

(2) $(|v\rangle, |w\rangle) = (|w\rangle, |v\rangle)^*$.

(3) $(|v\rangle, |v\rangle) \geq 0$ with equality if and only if $|v\rangle = 0$.

For example, $\mathbf{C}^n$ has an inner product defined by

$$((y_1, \dots, y_n), (z_1, \dots, z_n)) \equiv \sum_i y_i^* z_i = [y_1^* \dots y_n^*] \begin{bmatrix} z_1 \\ \vdots \\ z_n \end{bmatrix}. \quad (2.14)$$

- Vetores ortonormalizados

Vectors $|w\rangle$ and $|v\rangle$ are *orthogonal* if their inner product is zero. For example, $|w\rangle \equiv (1, 0)$ and $|v\rangle \equiv (0, 1)$ are orthogonal with respect to the inner product defined by (2.14). We define the *norm* of a vector $|v\rangle$ by

$$\| |v\rangle \| \equiv \sqrt{\langle v | v \rangle} . \quad (2.16)$$

Exercise 2.7: Verify that $|w\rangle \equiv (1, 1)$ and $|v\rangle \equiv (1, -1)$ are orthogonal. What are the normalized forms of these vectors?

- Algoritmo de Gram-Schmidt

Suppose $|w_1\rangle, \dots, |w_d\rangle$ is a basis set for some vector space V with an inner product. There is a useful method, the *Gram-Schmidt* procedure, which can be used to produce an orthonormal basis set $|v_1\rangle, \dots, |v_d\rangle$ for the vector space V . Define $|v_1\rangle \equiv |w_1\rangle / \| |w_1\rangle \|$, and for $1 \leq k \leq d - 1$ define $|v_{k+1}\rangle$ inductively by

$$|v_{k+1}\rangle \equiv \frac{|w_{k+1}\rangle - \sum_{i=1}^k \langle v_i | w_{k+1} \rangle |v_i\rangle}{\| |w_{k+1}\rangle - \sum_{i=1}^k \langle v_i | w_{k+1} \rangle |v_i\rangle \|}. \quad (2.17)$$

It is not difficult to verify that the vectors $|v_1\rangle, \dots, |v_d\rangle$ form an orthonormal set which is also a basis for V . Thus, any finite dimensional vector space of dimension d has an orthonormal basis, $|v_1\rangle, \dots, |v_d\rangle$.

Exercise 2.8: Prove that the Gram-Schmidt procedure produces an orthonormal basis for V .

There is a useful way of representing linear operators which makes use of the inner product, known as the *outer product* representation. Suppose $|v\rangle$ is a vector in an inner product space V , and $|w\rangle$ is a vector in an inner product space W . Define $|w\rangle\langle v|$ to be the linear operator from V to W whose action is defined by

$$(|w\rangle\langle v|) (|v'\rangle) \equiv |w\rangle \langle v|v'\rangle = \langle v|v'\rangle |w\rangle. \quad (2.20)$$

The usefulness of the outer product notation can be discerned from an important result known as the *completeness relation* for orthonormal vectors. Let $|i\rangle$ be any orthonormal basis for the vector space V , so an arbitrary vector $|v\rangle$ can be written $|v\rangle = \sum_i v_i |i\rangle$ for some set of complex numbers v_i . Note that $\langle i|v\rangle = v_i$ and therefore

$$\left(\sum_i |i\rangle\langle i| \right) |v\rangle = \sum_i |i\rangle\langle i|v\rangle = \sum_i v_i |i\rangle = |v\rangle. \quad (2.21)$$

Since the last equation is true for all $|v\rangle$ it follows that

$$\sum_i |i\rangle\langle i| = I. \quad (2.22)$$

$$A = I_W A I_V$$

$$= \sum_{ij} |w_j\rangle \langle w_j| A |v_i\rangle \langle v_i| \quad (2.24)$$

$$= \sum_{ij} \langle w_j| A |v_i\rangle |w_j\rangle \langle v_i|, \quad (2.25)$$

which is the outer product representation for A . We also see from this equation that A has matrix element $\langle w_j| A |v_i\rangle$ in the i th column and j th row, with respect to the input basis $|v_i\rangle$ and output basis $|w_j\rangle$.

Exercise 2.9: (Pauli operators and the outer product) The Pauli matrices (Figure 2.2 on page 65) can be considered as operators with respect to an orthonormal basis $|0\rangle, |1\rangle$ for a two-dimensional Hilbert space. Express each of the Pauli operators in the outer product notation.

2.1.5 Eigenvectors and eigenvalues

An *eigenvector* of a linear operator A on a vector space is a non-zero vector $|v\rangle$ such that $A|v\rangle = v|v\rangle$, where v is a complex number known as the *eigenvalue* of A corresponding to $|v\rangle$. It will often be convenient to use the notation v both as a label for the eigenvector, and to represent the eigenvalue. We assume that you are familiar with the elementary properties of eigenvalues and eigenvectors – in particular, how to find them, via the characteristic equation. The *characteristic function* is defined to be $c(\lambda) \equiv \det |A - \lambda I|$,

A *diagonal representation* for an operator A on a vector space V is a representation $A = \sum_i \lambda_i |i\rangle\langle i|$, where the vectors $|i\rangle$ form an orthonormal set of eigenvectors for A , with corresponding eigenvalues λ_i . An operator is said to be *diagonalizable* if it has a diagonal representation. In the next section we will find a simple set of necessary and sufficient conditions for an operator on a Hilbert space to be diagonalizable. As an example of a diagonal representation, note that the Pauli Z matrix may be written

$$Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} = |0\rangle\langle 0| - |1\rangle\langle 1|, \quad (2.29)$$

Exercise 2.11: (Eigendecomposition of the Pauli matrices) Find the eigenvectors, eigenvalues, and diagonal representations of the Pauli matrices X , Y , and Z .

Exercise 2.12: Prove that the matrix

$$\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \tag{2.31}$$

is not diagonalizable.

2.1.6 Adjoint and Hermitian operators

Suppose A is any linear operator on a Hilbert space, V . It turns out that there exists a unique linear operator $A^\dagger$ on V such that for all vectors $|v\rangle, |w\rangle \in V$,

$$(|v\rangle, A|w\rangle) = (A^\dagger|v\rangle, |w\rangle). \quad (2.32)$$

This linear operator is known as the *adjoint* or *Hermitian conjugate* of the operator A . From the definition it is easy to see that $(AB)^\dagger = B^\dagger A^\dagger$. By convention, if $|v\rangle$ is a vector, then we define $|v\rangle^\dagger \equiv \langle v|$. With this definition it is not difficult to see that $(A|v\rangle)^\dagger = \langle v|A^\dagger$.

In a matrix representation of an operator A , the action of the Hermitian conjugation operation is to take the matrix of A to the conjugate-transpose matrix, $A^\dagger \equiv (A^*)^T$, where the $*$ indicates complex conjugation, and T indicates the transpose operation. For example, we have

$$\begin{bmatrix} 1+3i & 2i \\ 1+i & 1-4i \end{bmatrix}^\dagger = \begin{bmatrix} 1-3i & 1-i \\ -2i & 1+4i \end{bmatrix}. \quad (2.34)$$

Operador Hermitiano: Um operador A é hermitiano ou autoadjunto se:

$$A = A^\dagger$$

An operator A whose adjoint is A is known as a *Hermitian* or *self-adjoint* operator. An important class of Hermitian operators is the *projectors*. Suppose W is a k -dimensional vector subspace of the d -dimensional vector space V . Using the Gram–Schmidt procedure it is possible to construct an orthonormal basis $|1\rangle, \dots, |d\rangle$ for V such that $|1\rangle, \dots, |k\rangle$ is an orthonormal basis for W . By definition,

Operador PROJETO

$$P \equiv \sum_{i=1}^k |i\rangle\langle i| \quad (2.35)$$

Operador NORMAL

An operator A is said to be *normal* if $AA^\dagger = A^\dagger A$. Clearly, an operator which is Hermitian is also normal. There is a remarkable representation theorem for normal operators known as the *spectral decomposition*, which states that an operator is a normal operator if and only if it is diagonalizable. This result is proved in Box 2.2 on page 72, which you should read closely.

Operadores UNITÁRIOS

A matrix U is said to be *unitary* if $U^\dagger U = I$. Similarly an operator U is unitary if $U^\dagger U = I$. It is easily checked that an operator is unitary if and only if each of its matrix representations is unitary. A unitary operator also satisfies $UU^\dagger = I$, and therefore U is normal and has a spectral decomposition. Geometrically, unitary operators are important because they preserve inner products between vectors. To see this, let $|v\rangle$ and $|w\rangle$ be any

- Operadores Positivos e Positivos-definidos (Pág. 71)

A special subclass of Hermitian operators is extremely important. This is the *positive operators*. A positive operator A is defined to be an operator such that for any vector $|v\rangle$, $(|v\rangle, A|v\rangle)$ is a real, non-negative number. If $(|v\rangle, A|v\rangle)$ is *strictly* greater than zero for all $|v\rangle \neq 0$ then we say that A is *positive definite*. In Exercise 2.24 on this page you will show that any positive operator is automatically Hermitian, and therefore by the spectral decomposition has diagonal representation $\sum_i \lambda_i |i\rangle\langle i|$, with non-negative eigenvalues λ_i .

Operador Positivo: Um operador A é positivo ou não-negativo se:

$$\langle \psi | A | \psi \rangle \geq 0$$

Operador Positivo Definido: Um operador A é positivo definido se:

$$\langle \psi | A | \psi \rangle > 0$$

Exercise 2.18: Show that all eigenvalues of a unitary matrix have modulus 1, that is, can be written in the form $e^{i\theta}$ for some real θ .

Como U é uma unitária normal
temos...

$$U = \sum_i \lambda_i |i\rangle \langle i| ; U^\dagger = \sum_i \lambda_i^* |i\rangle \langle i|$$

$$U^\dagger U = I$$

$$\sum_i \lambda_i^* |i\rangle \langle i| \left[\sum_j \lambda_j |j\rangle \langle j| \right] = I$$

$$\sum_i |\lambda_i|^2 |i\rangle \langle i| = I$$

$$\begin{pmatrix} |\lambda_1|^2 & 0 & \dots & 0 \\ 0 & |\lambda_2|^2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & |\lambda_n|^2 \end{pmatrix} = \begin{pmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{pmatrix}$$

$$|\lambda_i|^2 = 1$$

$$\rightarrow |\lambda_i| = 1$$

ou $\lambda_i = e^{i\theta}$, $\theta \rightarrow \text{REAL}$

Exercise 2.13: If $|w\rangle$ and $|v\rangle$ are any two vectors, show that $(|w\rangle\langle v|)^\dagger = |v\rangle\langle w|$.

Exercise 2.14: (Anti-linearity of the adjoint) Show that the adjoint operation is anti-linear,

$$\left(\sum_i a_i A_i\right)^\dagger = \sum_i a_i^* A_i^\dagger. \quad (2.33)$$

Exercise 2.15: Show that $(A^\dagger)^\dagger = A$.

Exercise 2.16: Show that any projector P satisfies the equation $P^2 = P$.

Exercise 2.17: Show that a normal matrix is Hermitian if and only if it has real eigenvalues.

Exercise 2.23: Show that the eigenvalues of a projector P are all either 0 or 1.

2.1.7 Tensor products

The *tensor product* is a way of putting vector spaces together to form larger vector spaces. This construction is crucial to understanding the quantum mechanics of multiparticle systems. The following discussion is a little abstract, and may be difficult to follow if you're not already familiar with the tensor product, so feel free to skip ahead now and revisit later when you come to the discussion of tensor products in quantum mechanics.

Suppose V and W are vector spaces of dimension m and n respectively. For convenience we also suppose that V and W are Hilbert spaces. Then $V \otimes W$ (read ‘ V tensor

W ’) is an mn dimensional vector space. The elements of $V \otimes W$ are linear combinations of ‘tensor products’ $|v\rangle \otimes |w\rangle$ of elements $|v\rangle$ of V and $|w\rangle$ of W . In particular, if $|i\rangle$ and $|j\rangle$ are orthonormal bases for the spaces V and W then $|i\rangle \otimes |j\rangle$ is a basis for $V \otimes W$. We often use the abbreviated notations $|v\rangle|w\rangle$, $|v, w\rangle$ or even $|vw\rangle$ for the tensor product

$|v\rangle \otimes |w\rangle$. For example, if V is a two-dimensional vector space with basis vectors $|0\rangle$ and $|1\rangle$ then $|0\rangle \otimes |0\rangle + |1\rangle \otimes |1\rangle$ is an element of $V \otimes V$.

By definition the tensor product satisfies the following basic properties:

- (1) For an arbitrary scalar z and elements $|v\rangle$ of V and $|w\rangle$ of W ,

$$z(|v\rangle \otimes |w\rangle) = (z|v\rangle) \otimes |w\rangle = |v\rangle \otimes (z|w\rangle). \quad (2.42)$$

- (2) For arbitrary $|v_1\rangle$ and $|v_2\rangle$ in V and $|w\rangle$ in W ,

$$(|v_1\rangle + |v_2\rangle) \otimes |w\rangle = |v_1\rangle \otimes |w\rangle + |v_2\rangle \otimes |w\rangle. \quad (2.43)$$

- (3) For arbitrary $|v\rangle$ in V and $|w_1\rangle$ and $|w_2\rangle$ in W ,

$$|v\rangle \otimes (|w_1\rangle + |w_2\rangle) = |v\rangle \otimes |w_1\rangle + |v\rangle \otimes |w_2\rangle. \quad (2.44)$$

What sorts of linear operators act on the space $V \otimes W$? Suppose $|v\rangle$ and $|w\rangle$ are vectors in V and W , and A and B are linear operators on V and W , respectively. Then we can define a linear operator $A \otimes B$ on $V \otimes W$ by the equation

$$(A \otimes B)(|v\rangle \otimes |w\rangle) \equiv A|v\rangle \otimes B|w\rangle. \quad (2.45)$$

to a convenient matrix representation known as the *Kronecker product*. Suppose A is an m by n matrix, and B is a p by q matrix. Then we have the matrix representation:

$$A \otimes B \equiv \left[\begin{array}{cccc} \overbrace{A_{11}B \quad A_{12}B \quad \dots \quad A_{1n}B}^{nq} \\ A_{21}B \quad A_{22}B \quad \dots \quad A_{2n}B \\ \vdots \quad \vdots \quad \vdots \quad \vdots \\ A_{m1}B \quad A_{m2}B \quad \dots \quad A_{mn}B \end{array} \right] \left. \vphantom{\begin{array}{c} A_{11}B \\ A_{21}B \\ \vdots \\ A_{m1}B \end{array}} \right\} mp. \quad (2.50)$$

In this representation terms like $A_{11}B$ denote p by q submatrices whose entries are proportional to B , with overall proportionality constant A_{11} . For example, the tensor product of the vectors $(1, 2)$ and $(2, 3)$ is the vector

$$\begin{bmatrix} 1 \\ 2 \end{bmatrix} \otimes \begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 1 \times 2 \\ 1 \times 3 \\ 2 \times 2 \\ 2 \times 3 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \\ 4 \\ 6 \end{bmatrix}. \quad (2.51)$$

The tensor product of the Pauli matrices X and Y is

$$X \otimes Y = \begin{bmatrix} 0 \cdot Y & 1 \cdot Y \\ 1 \cdot Y & 0 \cdot Y \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & -i \\ 0 & 0 & i & 0 \\ 0 & -i & 0 & 0 \\ i & 0 & 0 & 0 \end{bmatrix}. \quad (2.52)$$

Finally, we mention the useful notation $|\psi\rangle^{\otimes k}$, which means $|\psi\rangle$ tensored with itself k times. For example $|\psi\rangle^{\otimes 2} = |\psi\rangle \otimes |\psi\rangle$. An analogous notation is also used for operators on tensor product spaces.

Exercise 2.26: Let $|\psi\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$. Write out $|\psi\rangle^{\otimes 2}$ and $|\psi\rangle^{\otimes 3}$ explicitly, both in terms of tensor products like $|0\rangle|1\rangle$, and using the Kronecker product.

2.1.8 Operator functions

There are many important functions which can be defined for operators and matrices. Generally speaking, given a function f from the complex numbers to the complex numbers, it is possible to define a corresponding matrix function on normal matrices (or some subclass, such as the Hermitian matrices) by the following construction. Let $A = \sum_a a|a\rangle\langle a|$ be a spectral decomposition for a normal operator A . Define $f(A) \equiv \sum_a f(a)|a\rangle\langle a|$. A little thought shows that $f(A)$ is uniquely defined. This procedure can be used, for example, to define the square root of a positive operator, the logarithm of a positive-definite operator, or the exponential of a normal operator. As an example,

$$\exp(\theta Z) = \begin{bmatrix} e^\theta & 0 \\ 0 & e^{-\theta} \end{bmatrix}, \quad (2.56)$$

since Z has eigenvectors $|0\rangle$ and $|1\rangle$.

Seendo $\hat{A} = \sum_i a_i |a_i\rangle \langle a_i| \rightarrow$ DECOMPOSIÇÃO ESPECTRAL
DE UM OPERADOR
NORMAL

$$f(\hat{A}) \equiv \sum_i f(a_i) |a_i\rangle \langle a_i|, \text{ com } f(\hat{A}) \text{ ÚNICO:}$$

Exemplo: $f(\hat{Z}) = e^{\theta \hat{Z}}$, onde $\hat{Z} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$

SOLUÇÃO:

$\rightarrow$ LEMBRANDO QUE: $\hat{Z} = (+1) |0\rangle \langle 0| + (-1) |1\rangle \langle 1|$

Logo: $f(\hat{Z}) = e^{\theta (+1)} |0\rangle \langle 0| + e^{\theta (-1)} |1\rangle \langle 1|$

$$f(\hat{Z}) = e^{\theta} |0\rangle \langle 0| + e^{-\theta} |1\rangle \langle 1| = \begin{bmatrix} e^{\theta} & 0 \\ 0 & e^{-\theta} \end{bmatrix}$$

Exercise 2.35: (Exponential of the Pauli matrices) Let $\vec{v}$ be any real, three-dimensional unit vector and θ a real number. Prove that

$$\exp(i\theta\vec{v} \cdot \vec{\sigma}) = \cos(\theta)I + i \sin(\theta)\vec{v} \cdot \vec{\sigma}, \quad (2.58)$$

where $\vec{v} \cdot \vec{\sigma} \equiv \sum_{i=1}^3 v_i \sigma_i$. This exercise is generalized in Problem 2.1 on page 117.

Exercício 2.35

Seja $\vec{v}$ um vetor 3-dimensional real e unitário, (v_1, v_2, v_3) e θ um número real, prove que:

$$e^{i\theta(\vec{v} \cdot \vec{\sigma})} = \cos \theta \cdot I + i(\sin \theta) \cdot \vec{v} \cdot \vec{\sigma}$$

Dado: $\vec{v} \cdot \vec{\sigma} = \sum_{i=1}^3 v_i \sigma_i$

$$\hat{A} = \vec{v} \cdot \vec{\sigma} = v_1 \sigma_1 + v_2 \sigma_2 + v_3 \sigma_3$$

$$= \begin{pmatrix} 0 & v_1 \\ v_1 & 0 \end{pmatrix} + \begin{pmatrix} 0 & -iv_2 \\ iv_2 & 0 \end{pmatrix} + \begin{pmatrix} v_3 & 0 \\ 0 & -v_3 \end{pmatrix}$$

$$\hat{A} = \begin{pmatrix} v_3 & v_1 - iv_2 \\ v_1 + iv_2 & -v_3 \end{pmatrix}, \text{ observe } \hat{A} = \hat{A}^\dagger$$

$$\hat{A}^2 = \begin{pmatrix} v_3 & v_1 - iv_2 \\ v_1 + iv_2 & -v_3 \end{pmatrix} \begin{pmatrix} v_3 & v_1 - iv_2 \\ v_1 + iv_2 & -v_3 \end{pmatrix}$$

$$\hat{A}^2 = \begin{pmatrix} v_3^2 + v_1^2 + v_2^2 & 0 \\ 0 & v_3^2 + v_1^2 + v_2^2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}!$$

$$\Rightarrow (v_1 + i v_2)(v_1 - i v_2) = v_1^2 - i v_1 v_2 + i v_1 v_2 + v_2^2 = v_1^2 + v_2^2$$

$$(-i^2 v_2^2) \rightarrow$$

$$\Rightarrow v_3(v_1 - i v_2) + (v_1 - i v_2)(-v_3) = 0$$

Por isso, a exigência de o vetor $\vec{v}$ ter norma 1, isto é, $v_1^2 + v_2^2 + v_3^2 = 1$.

Então:

$$f(\vec{v} \cdot \vec{\sigma}) = e^{i\theta(\vec{v} \cdot \vec{\sigma})} = \cos\theta(\vec{v} \cdot \vec{\sigma}) + i \sin\theta(\vec{v} \cdot \vec{\sigma})$$

$$f(\vec{v} \cdot \vec{\sigma}) = \left[1 - \frac{\theta^2 \hat{A}^2}{2!} + \frac{\theta^4 \hat{A}^4}{4!} - \dots \right] + i \left(\theta \hat{A} - \frac{\theta^3 \hat{A}^3}{3!} + \frac{\theta^5 \hat{A}^5}{5!} - \dots \right)$$

$$f(\vec{v} \cdot \vec{\sigma}) = \left[1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \dots \right] \mathbb{I} + i \left[\theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \dots \right] \hat{A}$$

$$f(\vec{v} \cdot \vec{\sigma}) = \left[1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \dots \right] \mathbb{I} + i \left[\theta - \frac{\theta^3}{3!} \cancel{A^2 \mathbb{I}} + \frac{\theta^5}{5!} \cancel{A^4 \mathbb{I}} - \dots \right] \hat{A}$$

$$f(\vec{v} \cdot \vec{\sigma}) = (\cos \theta) \mathbb{I} + i(\sin \theta) \hat{A}$$

$$f(\vec{v} \cdot \vec{\sigma}) = (\cos \theta) \mathbb{I} + i(\sin \theta) (\vec{v} \cdot \vec{\sigma})$$

Fundamentos da Mecânica Quântica (MQ) para CQ: Os Postulados da MQ e Alguns Exemplos

Fundamentos da Mecânica Quântica

- Os postulados da mecânica quântica

Postulado - 1: Espaço estado do sistema físico

Postulate 1: Associated to any isolated physical system is a complex vector space with inner product (that is, a Hilbert space) known as the *state space* of the system. The system is completely described by its *state vector*, which is a unit vector in the system's state space.

$$|\psi\rangle = a|0\rangle + b|1\rangle,$$

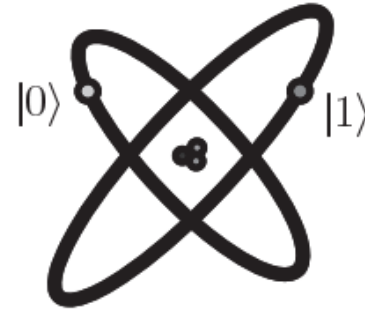


Figure 1.2. Qubit represented by two electronic levels in an atom.

where a and b are complex numbers. The condition that $|\psi\rangle$ be a unit vector, $\langle\psi|\psi\rangle = 1$, is therefore equivalent to $|a|^2 + |b|^2 = 1$. The condition $\langle\psi|\psi\rangle = 1$ is often known as the *normalization condition* for state vectors.

Postulado - 2: Evolução Temporal dos Sistemas Quânticos via Operadores Unitários

Postulate 2: The evolution of a *closed* quantum system is described by a *unitary transformation*. That is, the state $|\psi\rangle$ of the system at time t_1 is related to the state $|\psi'\rangle$ of the system at time t_2 by a unitary operator U which depends only on the times t_1 and t_2 ,

$$|\psi'\rangle = U|\psi\rangle. \quad (2.84)$$

$$|\Psi\rangle_{IN} \longrightarrow \boxed{U} \longrightarrow |\Psi\rangle_{OUT}$$

$$\alpha|0\rangle + \beta|1\rangle \longrightarrow \boxed{X} \longrightarrow \beta|0\rangle + \alpha|1\rangle$$

$$\alpha|0\rangle + \beta|1\rangle \longrightarrow \boxed{Z} \longrightarrow \alpha|0\rangle - \beta|1\rangle$$

$$\alpha|0\rangle + \beta|1\rangle \longrightarrow \boxed{H} \longrightarrow \alpha \frac{|0\rangle + |1\rangle}{\sqrt{2}} + \beta \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

Postulado - 3: Medidas Quânticas

Postulate 3: Quantum measurements are described by a collection $\{M_m\}$ of *measurement operators*. These are operators acting on the state space of the system being measured. The index m refers to the measurement outcomes that may occur in the experiment. If the state of the quantum system is $|\psi\rangle$ immediately before the measurement then the probability that result m occurs is

$$p(m) = \langle \psi | M_m^\dagger M_m | \psi \rangle ,$$

and the state of the system after the measurement is

$$|\Psi\rangle_{FINAL} = \frac{M_m |\psi\rangle}{\sqrt{\langle \psi | M_m^\dagger M_m | \psi \rangle}} .$$

The measurement operators satisfy the *completeness equation*,

$$\sum_m M_m^\dagger M_m = I . \quad (2.94)$$

The completeness equation expresses the fact that probabilities sum to one:

$$1 = \sum_m p(m) = \sum_m \langle \psi | M_m^\dagger M_m | \psi \rangle . \quad (2.95)$$

- **Medidas Projetivas:
Uma Classe Especial de Medidas**

Projective measurements: A projective measurement is described by an *observable*, M , a Hermitian operator on the state space of the system being observed. The observable has a spectral decomposition,

$$M = \sum_m m P_m , \quad (2.102)$$

where P_m is the projector onto the eigenspace of M with eigenvalue m . The possible outcomes of the measurement correspond to the eigenvalues, m , of the observable. Upon measuring the state $|\psi\rangle$, the probability of getting result m is

given by

$$p(m) = \langle \psi | P_m | \psi \rangle . \quad (2.103)$$

Given that outcome m occurred, the state of the quantum system immediately after the measurement is

$$\frac{P_m | \psi \rangle}{\sqrt{p(m)}} . \quad (2.104)$$

Projective measurements can be understood as a special case of Postulate 3. Suppose the measurement operators in Postulate 3, in addition to satisfying the completeness relation $\sum_m M_m^\dagger M_m = I$, also satisfy the conditions that M_m are orthogonal projectors, that is, the M_m are Hermitian, and $M_m M_{m'} = \delta_{m,m'} M_m$. With these additional restrictions, Postulate 3 reduces to a projective measurement as just defined.

Projective measurements have many nice properties. In particular, it is very easy to calculate average values for projective measurements. By definition, the average (see Appendix 1 for elementary definitions and results in probability theory) value of the measurement is

$$\mathbf{E}(M) = \sum_m m p(m) \quad (2.110)$$

$$\mathbf{E}(M) = \sum_m m p(m) \quad (2.110)$$

$$= \sum_m m \langle \psi | P_m | \psi \rangle \quad (2.111)$$

$$= \langle \psi | \left(\sum_m m P_m \right) | \psi \rangle \quad (2.112)$$

$$= \langle \psi | M | \psi \rangle. \quad (2.113)$$

This is a useful formula, which simplifies many calculations. The average value of the observable M is often written $\langle M \rangle \equiv \langle \psi | M | \psi \rangle$. From this formula for the average follows a formula for the standard deviation associated to observations of M ,

$$[\Delta(M)]^2 = \langle (M - \langle M \rangle)^2 \rangle \quad (2.114)$$

$$= \langle M^2 \rangle - \langle M \rangle^2. \quad (2.115)$$

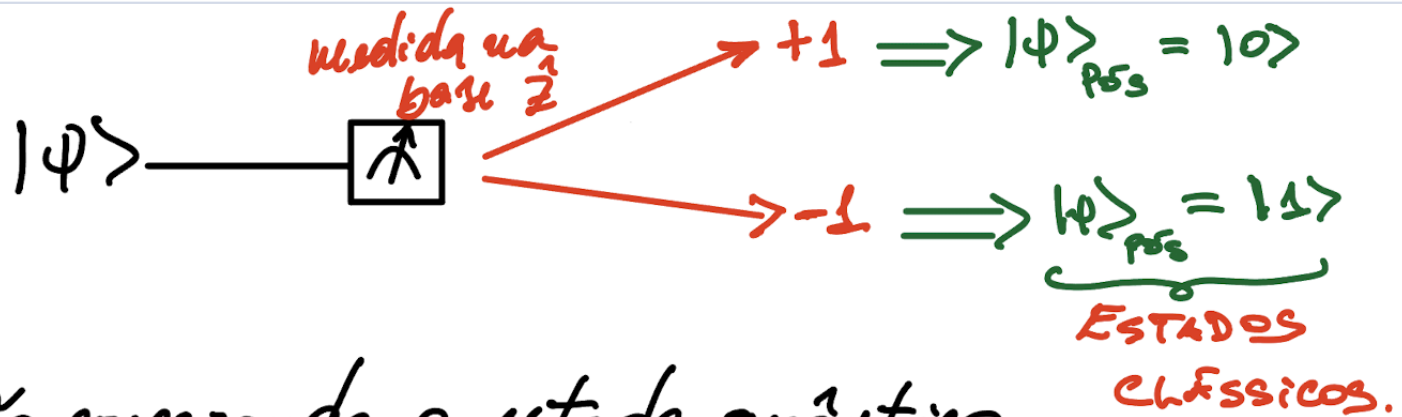
Exemplo:

Exemplo: Temos um observável $\hat{Z}$:

$$\hat{Z} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \text{ com os autovalores } \begin{cases} \hat{Z}|0\rangle = (+1)|0\rangle \\ \hat{Z}|1\rangle = (-1)|1\rangle \end{cases}$$

(+1) e (-1):

Então, supondo um estado quântico de um sistema físico de 2-níveis ($|0\rangle, |1\rangle$) (qubit): $|\psi\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$



Então preparando o estado quântico $|\psi\rangle$, várias vezes, da mesma forma, qual a probabilidade de se obter o estado clássico $|0\rangle$ ou $|1\rangle$?

Segundo o postulado da medida, temos que: $\hat{M} = \sum_i m_i P_i$ p/ $\hat{Z}$, temos:

$$\hat{Z} = (+1) P_1 + (-1) P_2$$

$$\hat{Z} = (+1)|0\rangle\langle 0| + (-1)|1\rangle\langle 1|$$

$$\begin{aligned}
 P(+1) &= \langle \psi | P_+ | \psi \rangle \\
 &= \langle \psi | 0 \rangle \langle 0 | \psi \rangle = |\langle 0 | \psi \rangle|^2 \\
 &= \left| \frac{\langle 0 | 0 \rangle + \langle 0 | 1 \rangle}{\sqrt{2}} \right|^2 \\
 &= \frac{1}{2}
 \end{aligned}$$

$$\begin{aligned}
 P(-1) &= \langle \psi | P_- | \psi \rangle \\
 &= \langle \psi | 1 \rangle \langle 1 | \psi \rangle = |\langle 1 | \psi \rangle|^2 \\
 &= \frac{1}{2}
 \end{aligned}$$

Estado Pós-MEDIDA, segundo o postulado da medida:

$$|\psi^{\text{pós}}\rangle = \frac{P_m |\psi\rangle}{\sqrt{P(m)}}$$

$$|\psi_{(+1)}^{\text{pos}}\rangle = \frac{(|0\rangle\langle 0|) |\psi\rangle}{\sqrt{\frac{1}{2}}} = \frac{|0\rangle\langle 0|0\rangle}{\frac{1}{\sqrt{2}}}$$

$$|\psi_{(+1)}^{\text{pos}}\rangle = |0\rangle$$

$$|\psi_{(-1)}^{\text{pos}}\rangle = \frac{P_z |\psi\rangle}{\sqrt{p_{(-1)}}} = \frac{|1\rangle\langle 1| \psi\rangle}{\sqrt{\frac{1}{2}}}$$

$$|\psi_{(-1)}^{\text{pos}}\rangle = |1\rangle$$

- Fase Relativa e Fase Global dos estados quânticos

* Fator de Fase Global dos estados quânticos

$$e^{i\theta}|\psi\rangle \quad |\psi\rangle$$

* Fator de Fase Relativa dos estados quânticos

$$\frac{|0\rangle + |1\rangle}{\sqrt{2}} \quad \text{and} \quad \frac{|0\rangle - |1\rangle}{\sqrt{2}}.$$

- Postulado - 4: Sistemas Compostos (n- sistemas)

Postulate 4: The state space of a composite physical system is the tensor product of the state spaces of the component physical systems. Moreover, if we have systems numbered 1 through n , and system number i is prepared in the state $|\psi_i\rangle$, then the joint state of the total system is $|\psi_1\rangle \otimes |\psi_2\rangle \otimes \cdots \otimes |\psi_n\rangle$.

Postulate 4 also enables us to define one of the most interesting and puzzling ideas associated with composite quantum systems – *entanglement*. Consider the two qubit state

$$|\psi\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}. \quad (2.132)$$

In	Out
$ 00\rangle$	$(00\rangle + 11\rangle)/\sqrt{2} \equiv \beta_{00}\rangle$
$ 01\rangle$	$(01\rangle + 10\rangle)/\sqrt{2} \equiv \beta_{01}\rangle$
$ 10\rangle$	$(00\rangle - 11\rangle)/\sqrt{2} \equiv \beta_{10}\rangle$
$ 11\rangle$	$(01\rangle - 10\rangle)/\sqrt{2} \equiv \beta_{11}\rangle$

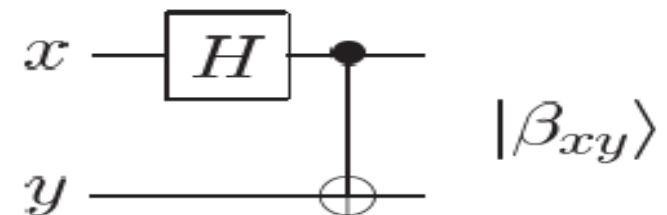


Figure 1.12. Quantum circuit to create Bell states, and its input–output quantum ‘truth table’.

2.2.9 Quantum mechanics: a global view

We have now explained *all* the fundamental postulates of quantum mechanics. Most of the rest of the book is taken up with deriving consequences of these postulates. Let's quickly review the postulates and try to place them in some kind of global perspective.

Postulate 1 sets the arena for quantum mechanics, by specifying how the state of an isolated quantum system is to be described. **Postulate 2** tells us that the dynamics of *closed* quantum systems are described by the Schrödinger equation, and thus by unitary evolution. **Postulate 3** tells us how to extract information from our quantum systems by giving a prescription for the description of measurement. **Postulate 4** tells us how the state spaces of different quantum systems may be combined to give a description of the composite system.

Referências

Para elaboração deste material de estudo os seguintes itens, figuras, exercícios e trechos de textos foram retirados da referência [1], capítulo 2. As demais referências foram citadas nas aulas expositivas e de laboratório.

[1] M. A. Nielsen and I. L. Chuang, Quantum Computation and Quantum Information, Cambridge University Press, (2000).

[2] Jack D. Hidary, Quantum Computing: An Applied Approach, Springer Nature Switzerland, (2019).

[3] Overview on quantum initiatives worldwide, <https://www.quireca.com/overview-on-quantum-initiatives-worldwide/>, Acesso em: 29.out.2021.

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